Yunpeng Meng

Curriculum Vitae

Department of Mathematics
Texas A&M University
College Station, TX 77843, USA
meng.yun@tamu.edu

Education

Ph.D.	**Texas A&M University**, College Station, TX, USA Mathematics, in progress Advisor: Tian Yang
M.S.	**Capital Normal University**, Beijing, China Mathematics
B.S.	**Tiangong University**, Tianjin, China Mathematics

Research Interests

Hyperbolic geometry and low-dimensional topology; the volume conjecture and quantum invariants of 3-manifolds; geometric triangulations and combinatorial Ricci flow; asymptotics of quantum $6j$-symbols; resurgence and Borel–Laplace resummation.

Papers and Preprints

[4] *On the volume conjectures for Dehn-filled 3-manifolds along fundamental shadow link complements with big volume*, with Baojun Wu, Tian Yang, and Yuxuan Yang.
Preprint, 2026.

[3] *Asymptotics of complex b-6j symbols*, with Tian Yang.
arXiv:2606.03653, 2026.

[2] *Hyperbolization and geometric decomposition of a class of 3-manifolds*, with Ke Feng and Huabin Ge.
arXiv:2503.07421, 2025.

[1] *On the volume conjecture for hyperbolic Dehn-filled 3-manifolds along the twist knots*, with Huabin Ge, Chuwen Wang, and Yuxuan Yang.
arXiv:2410.20489, 2024.

Teaching

Fall 2026	Grader, MATH 415H *Modern Algebra I* (Honors), Texas A&M University Instructor: Frank Sottile

References

Tian Yang, Ph.D. advisor
Associate Professor, Department of Mathematics
Texas A&M University, College Station, TX 77843, USA